
INMP441
Digital
microphone
I2S audio
STM32 Nucleo-
F446RE
Audio capture
and display
control
PCM audio frames
Status and start com
I2C status text
ESP32
Wi-Fi and HTTP
SSD1306 OLED
128 × 64
Chunked WAV upload
Recording command
FastAPI server
Note storage
and web
interface
Optional AI request
Transcription and
structured note
Gemini service
Record request
Playback and note res
User / web
browser